



Mae Fah Luang University

School of Applied Digital Technology



Software Engineering



Software Engineering

วิศวกรรมซอฟต์แวร์

IT School
Mae Fah Luang
University



Software Engineering (SE): We have updated the curriculum for the 2020 academic year to select the most in-demand skills in the market for our students. Below is the Software Engineering Framework, meticulously curated and selected by our team. This framework is the result of analysis by our faculty, industry experts, and alumni currently working in the field to ensure that the technology taught in each year is appropriate for both skill development and student revenue-generating models. (Here, what you learn can be applied immediately; use your skills to earn real income). Additionally, we reinforce knowledge in Emerging Technologies such as AI, ML, Big Data, Data Analytics, and Blockchain (*categorized in elective groups based on student aptitude or interest) to develop systems that are smarter and meet the needs of users and the job market.

Skills to be developed in each year:



- Year 1: The Newcomers** We learn the Agile software development process, logical thinking, and algorithms. This includes transforming problems into diagrams to be passed on to development teams. Students learn the principles of Object-Oriented Programming (OOP) and Java, which remains essential in current systems. We cover code management using Git for teamwork and popular industry tools like Eclipse and NetBeans. Students also learn website development using HTML5, CSS3, and JavaScript. During this year, learners can already begin taking programming or website projects ranging from hundreds to tens of thousands of baht.
- Year 2: Finding Your Feet** Students begin learning how to develop various systems, including Mobile Applications, Web Applications, and UX/UI design. We cover transforming user requirements into concise, clear Software Specifications, writing User Stories, and selecting appropriate technologies. Mobile development builds on previous coursework for both Android and iOS (based on interest through our supporting clubs). Web development utilizes frameworks such as Angular, Vue.js, Nuxt.js, React, and Spring Boot, alongside databases like MySQL, PostgreSQL, MongoDB, or Firebase Realtime Database. This ensures learners are "smart choosers" of tools and platforms. We also instill DevOps processes—Development, Build, Test, and Deploy—widely used in modern software firms, through both theory and practice. After Year 2, students can take on projects from thousands to tens of thousands of baht and begin publishing mobile apps to stores to generate sustainable passive income.



- **Year 3: Running Start** Students enter the phase of conceptualizing and developing their own projects using market-popular technologies (currently focusing on Docker, K8s, React, and Node). They learn to design software architecture, set up collaborative team environments, and manage medium-to-large scale projects. By creating Web APIs for large projects, students become confident in Full Stack Development. Additionally, they can explore alternative languages like Python for advanced technologies. By the end of this year, students reach a higher level of maturity, possessing the potential for self-learning and knowledge sharing. They can begin forming their own teams (approx. 3–5 people) to handle projects ranging from tens of thousands to hundreds of thousands of baht with close faculty consultation. Maintaining a "one app per month" mobile development habit can generate a passive income of approximately \$100–\$300 (3,000–9,000 THB) per month.
- **Year 4: Preparing for the Journey** This is the year of choices, regarding both internship locations and specialist career paths. Students can choose to join specialized interest groups such as Big Data, IoT (Internet of Things), Emerging Technologies (AI, ML, Blockchain), BI/BA (Business Intelligence/Business Analytics), or Enterprise Applications like SAP ERP. These groups engage in real-world projects, sharing knowledge in development, planning, and business negotiation, as well as accessing funding sources. With support from faculty, alumni, entrepreneur networks, investors, and university units, students are pushed to go further than they imagined. Upon graduation, qualified students can lead teams of 5–10 people and handle projects from tens of thousands to millions of baht. Continuous mobile app development can lead to a passive income of \$300–\$1,000 (9,000–30,000 THB) per month.

Courses Offered in Software Engineering



The elective groups are designed for balance, focusing on learning from the fundamentals of design, management, development, quality control, and architectural design, leading to specialized paths of interest. This prepares students to become professional software engineers or entrepreneurs.

Teaching and Learning Style

Instruction emphasizes both theory and practice. Students will gain hands-on experience through real-world projects and develop work for public release, maintaining the same standards as professional environments. For mobile applications, students will actually publish them to the App Store and Play Store, ensuring they gain genuine professional experience.

รูปแบบการเรียน การสอน

เรียนสนุก รู้ลึก ทำได้จริง

Workshop



ลงมือปฏิบัติ



พัฒนางานจริง



Extracurricular Learning Activities: Starting from their first year, freshmen will build their foundations through Code Camp activities. In the second year, students will practice systems development through Hackathon events. For the third and fourth years, students



will have the opportunity to exchange ideas with developers and software entrepreneurs through BarCamp activities.

กิจกรรมของสาขา วิศวกรรมซอฟต์แวร์



Studying in this field guarantees a great career and high pay.

These are sample job postings from leading IT websites showing that the average monthly salary after 2-3 years of experience ranges from 30,000 to 150,000 THB, depending on experience and ability. Software Engineering graduates are currently in high demand and highly sought after in the market because they possess a complete skill set, covering both development processes and software development skills.

SE - Jobs & Opportunities

฿ 20,000 – ฿150,000

QA Engineer (Multiple pos) - [Ad] Middle-Level, Full time ฿35,000-฿80,000 MAGE - กรุงเทพฯ Recruitment Quality Production	Junior Full Stack Developer Entry-Level, Full time ฿25,000-฿45,000 WISE - กรุงเทพฯ Bangkok Thailand Asia	NodeJS Backend Developer - [Ad] Senior-Level, Full time ฿60,000-฿150,000 Urban Mobility Tech Co.,Ltd. - กรุงเทพฯ กรุงเทพมหานคร Product	
Quality Assurance Q/A Middle-Level, Full time ฿30,000-฿80,000 Talad Holding - กรุงเทพฯ กรุงเทพมหานคร	Senior Full Stack Developer Senior-Level, Full time ฿40,000-฿80,000 WISE - กรุงเทพมหานคร Backend Api App App	Project Manager (2 roles) - [Ad] Senior-Level, Full time ฿70,000-฿100,000 MAGE - กรุงเทพฯ กรุงเทพมหานคร Communication Problem Solving Compliance Infrastructure Management	
Quality Assurance Engineer Middle-Level, Full time ฿25,000-฿80,000 True e-Logist - กรุงเทพมหานคร Recruitment Recruitment Product Recruitment	Frontend Developer Middle-Level, Full time ฿30,000-฿50,000 Verio - กรุงเทพมหานคร Java JavaScript Angular	Assistant Product Owner Middle-Level, Full time ฿25,000-฿80,000 WISESIGHT HQ (Thailand) - กรุงเทพฯ กรุงเทพมหานคร	
Senior QA / Senior Software Senior-Level, Full time ฿35,000-฿80,000 TSP Company - กรุงเทพมหานคร Java Recruitment Recruitment Recruitment	Frontend Developer Middle-Level, Full time ฿30,000-฿70,000 Oned - กรุงเทพมหานคร Angular React NodeJS	Business Development Entry-Level, Full time ฿30,000-฿55,000 Future Energy Cooperation Ltd. - กรุงเทพมหานคร Business Development Communication Marketing	

Studying SE means high market demand, great careers, and guaranteed high pay... or if you wish to pivot into becoming a



Startup founder or opening your own company, we have departments and faculty teams ready to provide full support.

Career Paths for Software Engineering Graduates

- **Software Analyst:** Base Salary 30,000 – 100,000 THB
- **Business Analyst:** Base Salary 30,000 – 100,000 THB
- **Programmer / Developer:** Base Salary 25,000 – 100,000 THB
- **Software Tester:** Base Salary 25,000 – 100,000 THB
- **Software Architect:** Base Salary 35,000 – 100,000 THB
- **Project Manager:** Base Salary 45,000 – 100,000 THB
- **Software Quality Assurance (SQA):** Base Salary 35,000 – 100,000 THB
- **Software Business Owner (CEO / Tech Startup):** Monthly Income 500,000 – 10,000,000 THB

Ask yourself: Is Software Engineering the right fit for you?

- Do you love learning new technologies and want to design and build them yourself?
- Do you want a high income while still enjoying system development and working with a team?
- Do you enjoy testing various systems and playing around with unique or unconventional software?
- Do you have ideas and want to transform them into products that can eventually generate revenue?
- Do you have the ambition to become a software entrepreneur with a target income of 1-10 million THB per month?

If you answered "Yes" to at least two of these questions, you are a great fit for this field. Everything starts with desire and internal drive; the rest—the learning process, senior students, faculty teams, and the environment—will help mold you to achieve your goals.



พิชัน SE รุ่น 1
CEO
Outsourcify

รุ่นพี่



พิก้า SE รุ่น 10
Software Engineer
DST

SE



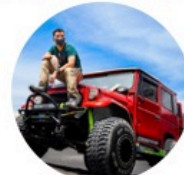
พิก้า SE รุ่น 7
Data Scientist
True Digital Group



พิก้า SE รุ่น 3
Mobile Engineer Manager
True Digital Group



พิก้า SE รุ่น 3
Agile Master & Team lead
G-Able



พิก้า SE รุ่น 2
Solution Architect
G-Able

We have a strong and tight-knit network of alumni from over 12 graduating classes. They are currently working in various fields and running their own companies both domestically and internationally, all ready to provide support and guidance to our students.

Programme

Degree Academic Programme Length of Programme

B.Eng. Software Engineering 4 Years

Programme's Philosophy

We teach students with international quality and academic knowledge in software engineering. They will gain expertise in the analysis, design, development, testing, and quality control of the software. The curriculum will instill students' ethical consciousness, professional ethics, and social responsibility.

Programme's Objectives

The curriculum aims to teach students who know to create software systems and can use commercial use or social development. They are ethical and uphold the code of conduct, can work together in a multicultural society. Nonetheless, students will have 21st-century



skills to keep pace with the changing technology. They can develop and build on their knowledge that integrates other related sciences.

Programme's Highlights

1. Learning the process of developing software and work systems from the small business level to the Enterprise level.
2. Covering Mobile Application Development, Web Application, Service Oriented Architecture, and Software Technology Stack.
3. Activities to promote relationships between teachers, seniors, and juniors in the SE Hackathon event. Company and technology organization at Bar Camp Chiang Rai event
4. Pre Internship that allows potential students to train with leading companies
5. Special Topic that is open to support the rapidly changing world of technology.

Career Opportunities

1. Software Engineer
2. Programmer/Developer
3. System Analyst /Designer
4. Requirements Engineer
5. Software Quality Assurance
6. Software Process Improvement Engineer
7. Software Tester
8. Software Architect
9. System Integrator
10. Software Project Manager

Programme Learning Outcomes: PLOs

- PLO 1: Honesty, self-responsibility, consideration for common interests, and professional ethics
- PLO 2: Basic knowledge of software development processes, software manufacturing methods, structural design, software



architecture, quality control, and software development project management.

- PLO 3: Choose the solution to the problem using software engineering knowledge systematically and appropriately.
- PLO 4: Work with others in a multicultural society
- PLO 5: Have numerical analysis skills Use of communication tools and information technology
- PLO 6: Build software or systems that keep pace with changing technology and can use for commercial purposes or social development

Tuition and Fees

- 39,000 Baht per semester
- 312,000 Baht per programme

Programme Structure

- **Programme Structure**
 - Bachelor of Engineering Programme in Software Engineering
 - **Total credits: 125 credits (4-year Bachelor's Degree)**

1. General Education Course 30 credits

2. Specific Requirement Course 90 credits

2.1 Core Course 9 credits

2.2 Major Required Course 63 credits

2.3 Co-operative Education 9 credits

2.4 Major Elective Course 9 credits

3. Free Elective Course 6 credits

(Refer to Bachelor of Science Program in Software Engineering Year 2022)

Additional Program Details

TQF.2



Bachelor's Degree

Digital and Communication Engineering

Computer Engineering

Digital Technology for Business Innovation

Software Engineering

Multimedia Technology and Animation



School of Applied Digital Technology
Mae Fah Luang University

Contact Us

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ระเบิดไอดีสุดล้ำ! 🚀 ประมวลภาพบรรยากาศ
ความสนุกจากงาน MediHack Junior 2026 🏆💻
สำนักวิชาเทคโนโลยีดิจิทัลประยุกต์ (Applied Digital
Technology) ได้จัดกิจกรรม MediHack Junior